

TOPICS FOR FINAL EXAM, MATH 55 SPRING 2026

You must know all definitions and proofs presented in class, as well as proofs that I told you to read in the book. You must write all answers in complete English sentences.

Chapter 1. Main topics:

- Propositional logic
- Propositional equivalences
- Predicates and quantifiers
- Nested quantifiers and negation of quantifiers
- Rules of inference (don't need to memorize names)
- Basic proof techniques: direct, contrapositive, contradiction, cases.

Types of problems

- Write truth tables for propositions
- Determine when two propositions are equivalent
- Determine when proposition is a tautology
- Turn English statements into propositions and vice-versa
- Determine truth values of propositional functions
- Use rules of inference to build valid arguments
- To identify logical fallacies in arguments

Chapter 2. Main topics:

- Elements, sets, subsets; know difference between elements and sets
- Set operations (power set, Cartesian product, union, difference, complement)
- Functions (injective, surjective, bijective; inverses and compositions)
- Cardinality, countable and uncountable sets, diagonalization argument

Types of problems:

- Prove set identities
- Determine (with proof) whether a function is injective, surjective, bijective
- Determine (with proof) whether a set is countable or uncountable

Chapter 4. Main topics:

- Basic properties of divisibility, primes and composites
- Existence of infinitely many primes
- The Well-Ordering Principle
- Division algorithm
- Euclidean algorithm and GCD
- Bezout's theorem and strong version
- Euclid's lemma
- Fundamental theorem of arithmetic
- Basic properties of arithmetic modulo m
- Existence of inverses when a, m are relatively prime
- Fermat's little theorem

Types of problems:

- Prove simple statements about divisibility
- Prove more sophisticated statements, possibly using the named theorems listed above
- Convert numbers into binary and back again
- Use the Euclidean algorithm to find the gcd and Bezout coefficients $sa + tb = 1$
- Find the inverse of a mod m

- Use Fermat's Little Theorem to compute large powers modulo m

Chapter 5. Main topics:

- Induction (key examples: proofs of inequalities and identities, divisibility, statements about sequences defined recursively, statements about unions and intersections of n sets).
- Strong induction (key examples: tiling with dominos, making monetary amounts, outcomes of games such as the one we played in class, induction with graphs)
- Well-ordering principle (and why it implies that induction works)
- Recursive definitions (key examples: Fibonacci, Euclidean Algorithm)

Types of problems:

- Proofs using induction and its variants, including proofs of statements about topics in other chapters.

Chapter 6. Main topics

- Cardinality, product and sum rules
- Bijections, many to one maps, division rule
- Permutations and combinations, with and without repetitions
- Combinatorial proofs
- Pigeonhole Principle (including Generalized Pigeonhole Principle, and more sophisticated applications such as: must have 3 friends or 3 strangers in a group of 6).
- Binomial coefficients, Binomial Theorem, Pascal's identity
- Inclusion-exclusion

Types of problems

- Basic counting using product, sum, and division rules (by defining a process for constructing the objects you want to count)
- Proofs using the pigeonhole principle
- Counting permutations and combinations with and without repeated objects
- Proofs involving binomial coefficients, including combinatorial proofs (i.e., counting the same thing in two different ways)

Chapter 8. Main topics:

- Recurrence relations,

Types of problems:

- expressing the solution to a counting problem as a recurrence

Graph Theory (Chapter 10). Main topics

- Basic definitions: edge, vertex, (simple) graph, multigraph, adjacent, incident, endpoint, neighborhood, degree, path, circuit, simple path/circuit, Euler circuit, connected, subgraph, union, k -colorable connected component, tree, acyclic, leaf, k -coloring, chromatic number, isomorphism
- Special types of graphs (complete, cycle, hypercube, etc)
- Connected components of a graph
- Handshaking theorem
- 2-colorable iff no odd circuit
- Chromatic number at most one plus maximum degree
- Euler circuit iff connected and even degrees
- Basic properties of trees and spanning trees (unique path, number of edges, existence of leaves)

Types of problems:

- Determine whether a given graph is 2-colorable, has an Euler circuit, is connected; find a coloring of a given graph or show that none exists
- Draw a particular graph.
- Determine whether a degree sequence can be realized by a simple graph
- Proofs relating notions such as degree, partitions, connectivity, 2-colorability, chromatic number, and Euler circuits, possibly requiring application of the above theorems

- Proofs using induction (where $P(n)$ will depend on either the number of vertices or the number of edges, and a smaller graph is obtained by deleting an edge or vertex)

Directed graphs and graphs with loops are not covered in this course.

Chapter 7. Main topics

- Probability (experiment, sample space, event, outcome, probability distribution)
- Conditional probability, independent events, law of total probability (key examples: Monty Hall)
- Bayes' Theorem
- Random variables, expected value, linearity of expectation, algebra with random variables
- Relationship between events and random variables
- Coin flips (Bernoulli Trials), the binomial distribution, coupon collector example
- Independent random variables
- Variance and Chebyshev's inequality

Types of problems

- Be able to clearly specify the experiment, sample space, probabilities, and events given a word problem
- Compute probabilities and conditional probabilities using basic definitions and counting (e.g. for poker hands or coin flips)
- Compute probabilities using the law of total probability and Bayes' theorem.
- Determine whether events and random variables are independent
- Compute the expectation and variance of random variables
- Apply linearity of expectation
- Apply Chebyshev's inequality and Markov's inequality
- Proofs of basic properties of expected value and variance, including variance of sums of independent random variables
- Analysis of random graphs using linearity of expectation
- Proofs using the probabilistic method

Geometric random variables will not be on the exam.